

Airborne mutagens produced by frying beef, pork and a soy-based food.



Airborne cooking by-products from frying beef (hamburgers), pork (bacon strips) and soybean-based food (tempeh burgers) were collected, extracted, tested for mutagenicity and chemically analysed. The fumes generated by frying pork and beef were mutagenic, with 4900 and 1300 revertants/g of food cooked, respectively. No mutagenicity was detected in fumes from frying tempeh burgers. Bacon fried to a well-done but non-charred state was eight times more mutagenic in a microsuspension Ames/Salmonella test (TA98 with S-9) than hamburgers and about 350 times more mutagenic than tempeh burgers. Among food samples cooked to a well-done, non-charred state, bacon strips had almost 15-fold more mass (109.5 ng/g) than that of the beef, whereas no heterocyclic amine (HCA) was detected in the fried tempeh burgers. 2-Amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) was the most abundant HCA, followed by 2-amino-3,8-dimethylimidazo[4,5-f]quinoxaline (MeIQx) and 2-amino-3,4,8-trimethylimidazo[4,5-f]quinoxaline (DiMeIQx). No 2-amino-9H-pyrido[2,3-b]indole (A α C) was detected in the food samples fried at about 200 degrees C, although it was present in the collected airborne products. The total amounts of HCAs in the smoke condensates were 3 ng/g from fried bacon, 0.37 ng/g from fried beef and 0.177 ng/g from fried soy-based food. This study indicates that cooks are potentially exposed to relatively high levels of airborne mutagens and carcinogens and that long-term sampling inside restaurants and kitchens may be warranted in order to assess the potential risk of prolonged exposure.